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Do Digital Visual Aids Improve Skin Self-Examination Behaviours? A Systematic Review

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Abstract

Digital visual aids like smartphone applications can potentially increase uptake and maintenance of Self-Skin Examination (SSE) behaviours, which is an essential secondary prevention strategy for early detection of skin cancer. This review aimed to understand whether digital visual aids with lesion monitoring functions improve SSE behaviours without negatively impacting health outcomes. A systematic search was performed using six databases for relevant literature published before June 2025. Studies were included if they used digital interventions that could monitor skin lesions and reported outcomes on SSE behaviours or their antecedents, such as SSE knowledge, attitudes and intentions. Data were analysed following an adapted Technological Acceptance Model framework. Twenty studies conducted in the United States, the United Kingdom, or Australia involving 19 to 1,000 participants were included. Nineteen studies were randomised controlled trials (RCTs) or nested within RCTs, with one non-nested cross-sectional study. Interventions were delivered by smartphone and/or tablet applications, or websites. Studies found digital visual aids led to improved knowledge ($n = 6$), perceived usefulness ($n = 5$), ease of use ($n = 7$), attitudes towards SSE ($n = 13$), as well as intention to perform regular SSE ($n = 5$). Overall, there was a positive impact on health outcomes, with only two studies reporting an increase in anxiety and worry following the intervention. Three of seven studies demonstrated an increased likelihood of detecting malignant skin lesions when monitoring using digitally aided SSE. There is potential for digital visual aids to improve SSE behaviours. However, it is important to address barriers to the use of digital visual aids for SSE and to improve their integration with clinical care.

Keywords

digital health, mHealth, self-surveillance, melanoma, skin cancer

Introduction

The incidence of melanoma and other skin cancers is increasing globally, and the 5-year survival rate for those diagnosed with stage IV melanoma is only 26% (Australian Institute of Health and Welfare, 2024; Cancer Australia, n.d.). Preventive health strategies, early diagnosis and timely excision of melanoma are critical for improving patient outcomes (Guy & Ekwueme, 2011). Currently, in most countries around the world, melanoma detection relies on opportunistic screening by health care providers (HCPs) and vigilant self-monitoring by patients. For melanoma survivors, clinician skin examination (CSE) is generally recommended every 4 to 6 months for several years following the primary diagnosis to monitor for recurrence (Gordon et al., 2022; Thompson et al., 2012).

Skin self-examination (SSE) is a low-cost but potentially high-impact method of melanoma detection. SSE involves the inspection of one's own skin for suspicious or changing moles and is recommended for individuals at high risk of skin cancer (Cancer Council Australia, n.d.-a, n.d.-b). Studies report that 57% of first melanoma and 40% to 67% of melanoma recurrences are identified by patients or their family members (Berger et al., 2017; Bhurosy et al., 2020; Brady et al., 2000; Lee et al., 2017; Meyers et al., 2009). However, SSE is often limited by poor technique, lack of knowledge on what to look

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for and inconsistent practice (Körner et al., 2013), limitations which digital visual aids may help to address (Kong et al., 2021). We define digital visual aids as electronic visual tools that can assist in SSE practice. There are four main ways digital aids have been shown to assist SSE: (a) delivering education, (b) monitoring lesions, (c) accessing HCPs for advice about a skin lesion and (d) automated diagnosis with artificial intelligence (AI) (Kong et al., 2021). The British Association of Dermatologists (BAD) does not currently recommend the use of AI-based digital aids due to accuracy concerns and lack of randomised trial evidence (British Association of Dermatologists, 2025; Hussain et al., 2026). However, the monitoring functions of digital aids may support existing self-surveillance strategies by prompting earlier intervention and optimising intervals between clinician visits (Janda et al., 2004). The primary aim of this study is to evaluate whether digitally aided SSE leads to behaviour change by improving knowledge, attitudes, intention and performance of SSE. The secondary aim is to determine the impact of these aids on health outcomes.

Method

Search Strategy

A systematic search for relevant literature was performed in August 2022 and repeated in January 2024 and June 2025 on Embase, CINAHL, Cochrane Library, Scopus, ERIC and PubMed databases. Search terms were selected with librarian guidance and were based on the study aim (Supplemental Table S1). A protocol of the systematic review was registered under ID CRD42022335729 on the International Prospective Register of Systematic Reviews (PROSPERO) and reporting follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (National Institute for Health and Care Research, n.d.). Ethical approval was not required for this systematic review of previously published data.

Study Eligibility and Selection

Studies were eligible for inclusion if they used a digital visual aid to help people monitor skin lesions and assessed outcomes related to SSE behaviours or their antecedents. Participants had to be 18 years or older. The search was limited to literature published between January 2000 and June 2025 and in English. Literature with no full text available, theses, posters or conference abstracts were excluded from this review. Two reviewers independently screened the literature titles and abstracts using Covidence software (Veritas Health Innovation, n.d.), followed by a full-text review to assess eligibility. Disagreements were resolved through discussions between the reviewers. Conflicts that could not be resolved were escalated to a third research team member for a final decision.

Extracting Primary and Secondary Outcomes

Using a modified version of the Technology Acceptance Model (TAM), the primary outcome was to evaluate whether digital aids would lead to improvements in SSE knowledge, attitudes, intentions and behaviours (Horsham et al., 2020). SSE behaviours were defined as SSE frequency or thoroughness. The secondary outcome was to assess the impact of the digital visual aids on patient health-related outcomes, including anxiety, worry and skin cancer detection, as well as broader health system implications, including interactions with HCPs.

The TAM is a well-established framework that identifies key determinants of technology acceptance and explains how these factors influence and support subsequent behavioural change. The conceptual application of TAM to this study context is presented in Figure 1.

Study details and outcome data were extracted from each article, categorised according to the TAM stages and checked for consensus across reviewers. Conflicts were resolved through discussion or, where necessary, arbitrated by a third team member.

Quality Assessment

The quality of randomised controlled trials (RCTs) was assessed using the Cochrane Risk of Bias tool (Flemyng et al., 2020), where each domain was rated as either high, fair/some concerns, or low risk of bias. The National Institute of Health's (NIH) Before–After (Pre–Post) Studies with No Control Group quality assessment tool was used to assess the quality of non-RCT studies, and studies nested within RCTs (National Institute of Health, n.d.). The domains assessed in the NIH Quality Assessment Tool were rated as poor, fair or good, and the overall trend of these ratings denoted the final evaluated risk of bias of the study. Risk of bias was checked for consensus across reviewers.

Results

A total of 1,477 studies were imported to Covidence for screening, from searches conducted up to June 2025. After duplicates were identified and removed, 670 studies were screened based on their title and abstract. Of these 670 studies, 109 progressed to full-text screening (Figure 2). One additional study was identified through manual search, as it reported relevant results of a screened study. Twenty studies were included in this review, with most studies excluded for irrelevant intervention (i.e. not digital visual aids or no monitoring function; $n = 53$), or ineligible publication type ($n = 14$).

Only four studies were classified as "low risk" of bias (Geller et al., 2023; Manne et al., 2021; Manne, Heckman, et al., 2022; Robinson et al., 2021), with the remaining studies classified as "at risk" of bias, mostly due to limitations in

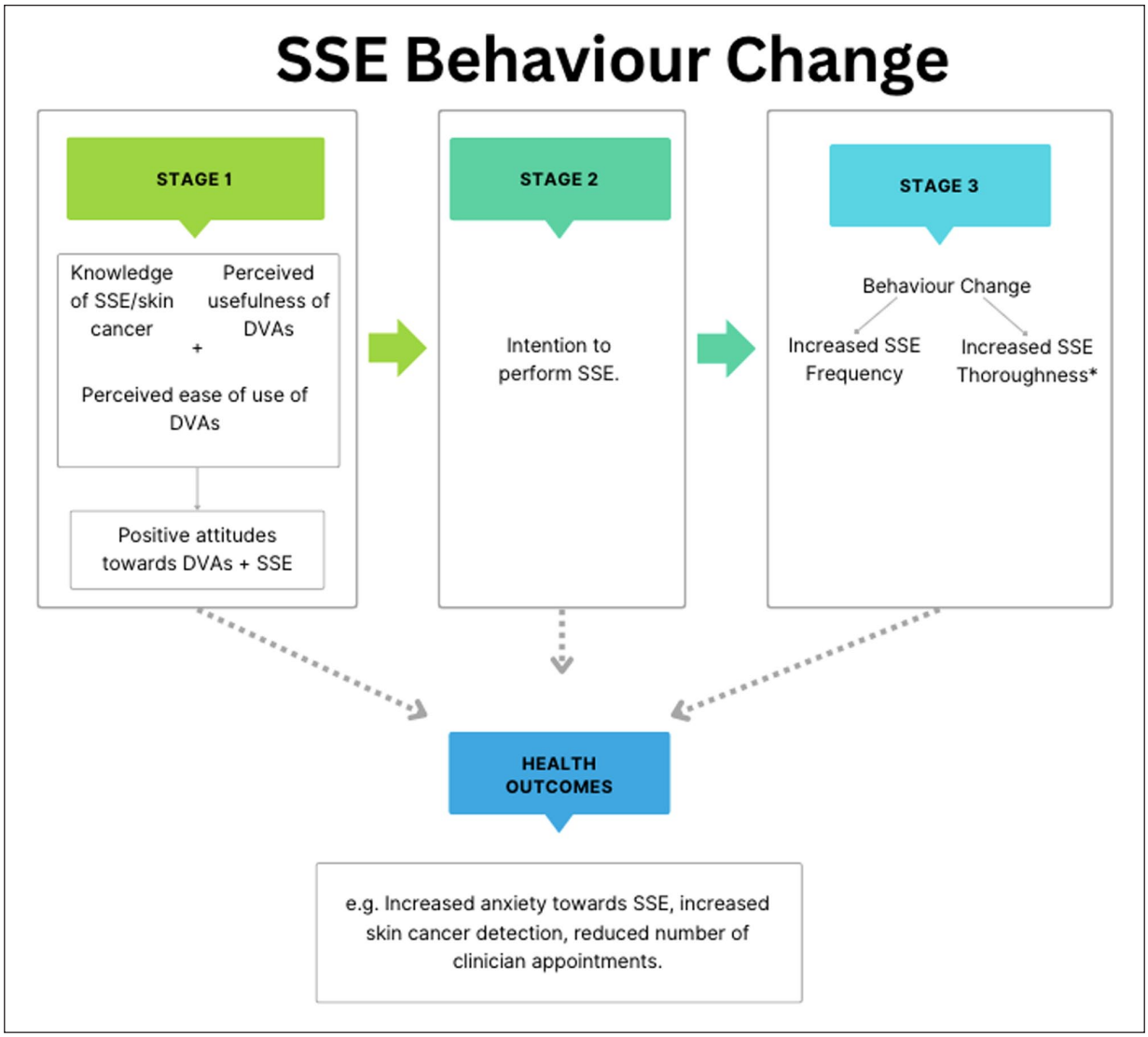


Figure 1. Adapted Technology Acceptance Model (TAM) demonstrating the stages of skin self-examination (SSE) behaviour change with acceptance of digital visual aids following intervention exposure.
[a]Increased SSE thoroughness is defined by increased number of bodily parts examined, increased use of partner assistance or mirror usage.

how outcomes were measured or unreported outcome data. For nested studies, bias concerns stemmed from unclear selection criteria, small sample sizes with loss to follow-up of over 20%, and unspecified or inconsistent outcome measures (Supplemental Tables S2 and S3).

Study Characteristics

Key characteristics of the 20 studies are reported in Table 1. The majority of included studies were RCTs, or nested within RCTs (n = 19/20), with eight of those reporting only on the intervention arm of the RCTs. In addition, this review included one non-nested cross-sectional study (Hampton et al., 2020). Eight studies were conducted in the United States, seven in the United Kingdom and five in Australia. Sample sizes ranged from 19 to 1,000, with nine studies having a minimum of 100 participants. Nine studies reported a participant mean age of 50 years or older, and none focused on young people alone. Eight studies included participants from rural or remote areas; the remaining 12 did not specify the locality of their participants. Most studies recruited some combination of melanoma survivors who had either been treated for early melanoma in the last 5 to 10 years (n = 14/20), surgically treated for stages I–III melanoma in the last 2 years (n = 5/20), or had high-risk features for skin cancer but no melanoma history (n = 8/20). Only one study did not specify the skin cancer risk category of their participants (Marek et al., 2018a).

Intervention Characteristics

Interventions were delivered using smartphone applications (n = 6/20), smartphone applications with tablets provided (n

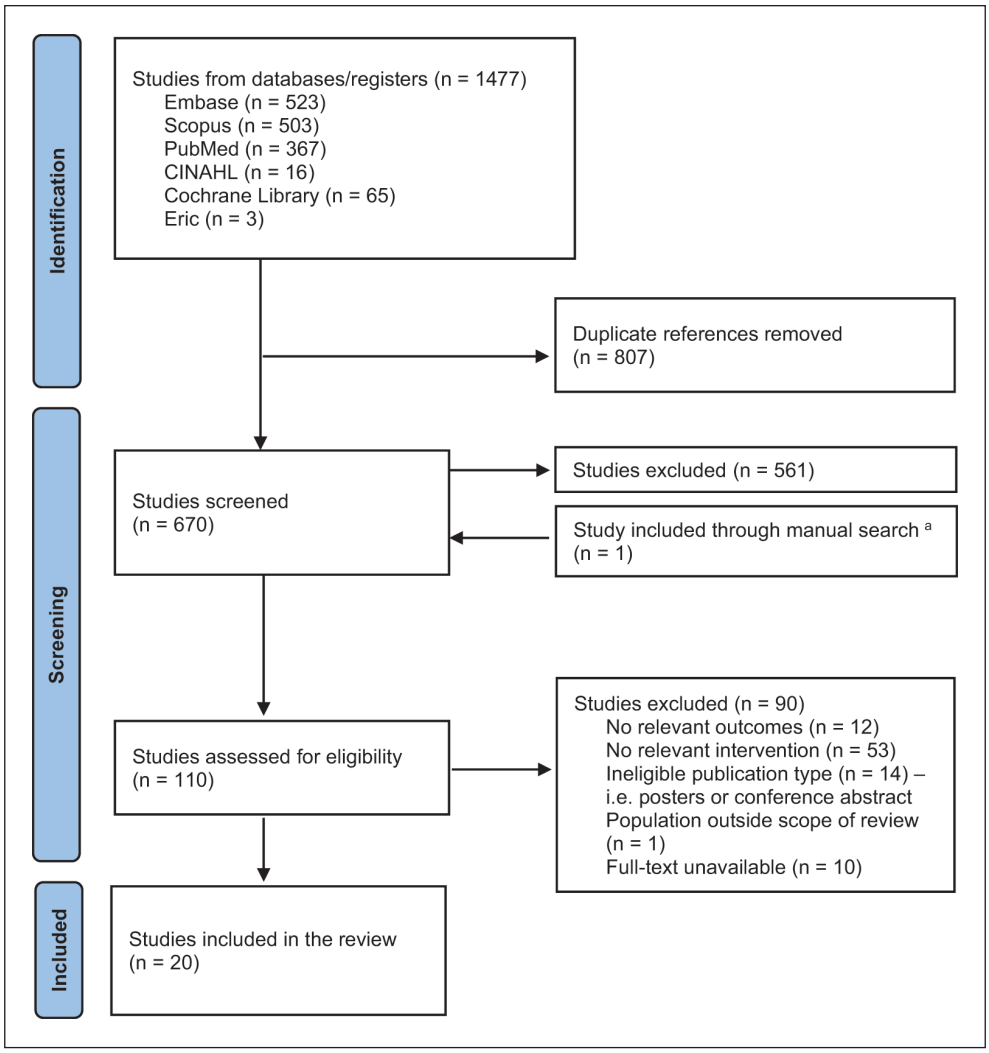


Figure 2. PRISMA diagram presenting the search strategy and study selection.
[a]One additional study was identified through manual search, as it reported the relevant results of a screened study.

= 4/20), tablets only with in-built lesion tracking software (n = 2/20), smartphone applications with dermatoscope attachments (a hand-held device used to magnify skin lesions; n = 3/20), websites (n = 3/20), smartphone applications with dermatoscope attachments and websites (n = 1/20), or dermatoscope attachments with a website (n = 1/20). To monitor lesions, 12 studies had integrated photo storage functions within the intervention, four studies employed the participants' usual smartphone camera, and four studies used body mole maps for participants to mark their lesions (Manne et al., 2021; Manne, Heckman, et al., 2022; Manne, Marchetti, et al., 2022; Turrisi et al., 2015). A total of 15 studies incorporated reminders into their interventions to encourage repeated SSE. Training for the intervention was delivered face-to-face in three studies, was integrated in the application in 14 studies (e.g., via instructions, video demonstration or structured checklist), and 15 studies provided participants with materials (e.g., brochures or workbooks).

Outcome Measures

Study outcomes are presented in Table 1 and in Supplemental Table S4. Among the selected studies, some evaluated the same digital platform, including the Achieving Self-directed Integrated Cancer Aftercare (ASICA) app (n = 7/20) and mySmartSkin (n = 3/20). For ASICA, four papers were derived from the same pilot RCT but examined different outcomes or aspects of use, such as feasibility, barriers and facilitators and user experiences. For mySmartSkin, two studies were derived from the same RCT, with one study examining the effects of platform use and the other assessing potential moderators.

Most studies used self-administered surveys for outcome measurements (n = 16/20). Half used surveys only, while the other half supplemented surveys with additional data sources. The remaining four studies used interviews and/or app data metrics. All studies measured outcomes pertaining to at least one stage of the proposed behaviour change model (Figure 1).

Table 1. Study Characteristics.

Author, year	Country	Trial and study type	Sample size (*n*), % male	Mean age or range	Includes rural or remote region	Melanoma risk/ skin cancer risk	Intervention format	Period	Includes training	Outcome measure tool/ methods	Overall SSE behavioural outcome[a]	Health outcome
Ackermann et al. (2022)	Australia	MELSELF[b] ASICA[c,d] Pilot RCT study	100, 53%	59[e]	NS	Treated for cutaneous melanoma[f]	Smartphone application, dermatoscope attachment and website	6 months	✓	Self-administered survey at multiple time points and interviews	Positive INT was more likely to perform SSE (OR = 3.5)[g]	No difference in psychological outcomes (fear of recurrence, depression, anxiety and stress)[h]
Dennis et al. (2015)	The United Kingdom	ASICA[c] Pilot qualitative study	20, 45%	(37–83)[h]	Yes	Treated for cutaneous melanoma[f] past 9 years	Smartphone application and tablet	6 months	✓	App data and interviews	Positive Patients were encouraged to perform skin checks more often	NIL
Drabarek et al. (2022)	Australia	MELSELF [b] Nested qualitative study	20, 60%	57	Yes	Treated for cutaneous melanoma[f] duration not mentioned	Smartphone application and dermatoscope attachment	6 months	*X*	App data and interviews	Positive 80% of participants in the INT used the intervention tools	Reported increased anxiety or worry
Geller et al. (2023)	The United States	ASK[i] RCT study	728, 47%	(30–65)[e]	NS	No personal history of skin cancer	Dermatoscope attachment and website	Up to 18 months	✓	Self-administered survey at multiple time points and medical records	Positive Increase in rates of CSE (RR = 1.89) and SSE (RR = 1.76) from baseline to 12 months[h]	NIL
Hampton et al. (2020)	The United Kingdom	MySkinSelfie Cross-sectional study	19, NS	46 (22–72)	NS	Includes those with skin cancer surgery or monitoring	Smartphone application only	3 months	*X*	Self-administered survey after intervention	Positive Users reported positive attitude towards intervention and high perception of ease of use	NIL
Horsham et al. (2020)	Australia	Survey, part of RCT study	98, 32%	42 (19–73)	NS	Have at least two risk factors for skin cancer[j]	Smartphone application only	3 months	✓	Self-administered survey after intervention and app data	Negative Acceptance towards digital aid decreased significantly[g] by 0.21 points	Most participants reported not feeling anxious during SSE (75%, *n* = 98)[h]
Janda et al. (2020)	Australia	RCT study	199, 29%	41 (18–73)	NS	Have at least two risk factors for skin cancer[j]	Smartphone application and dermatoscope attachment	3 months	*X*	Self-administered survey and app data	Fewer participants completed all SSE[g] (INT = 83% vs CG = 95%)	NIL
Kong et al. (2020)	Australia	Qualitative study, part of RCT	28, 46%	42 (18–60)	NS	Have at least two risk factors for skin cancer[h]	Smartphone application and dermatoscope attachment	2 months	*X*	Self-administered survey	NIL	No change or decrease in anxiety or worry

(continued)

Table 1. (continued)

Author, year	Country	Trial and study type	Sample size (*n*), % male	Mean age or range	Includes rural or remote region	Melanoma risk/ skin cancer risk	Intervention format	Period	Includes training	Outcome measure tool/ methods	Overall SSE behavioural outcome[a]	Health outcome
Manne et al. (2021)	The United States	mySmartSkin RCT study	441, 51%	61 (18–89)	NS	Diagnosed with stages 0–III melanoma	Website	12 months	✓	Self-administered survey at multiple time points and medical records	Positive Significantly[g] more participants in the INT group reported conducting thorough SSE at each follow-up	NIL
Manne, Heckman, et al. (2022)	The United States	mySmartSkin RCT study	441, 51%	61 (18–89)	NS	Diagnosed with stages 0–III melanoma	Website	12 months	✓	Self-administered survey at multiple time points	Positive Participants who viewed the digital aid more, were more likely[g] to do a thorough SSE at each follow-up	NIL
Manne, Marchetti, et al. (2022)	The United States	mySmartCheck[k] RCT study	116, 30%	51[e]	NS	Increased risk for melanoma[j]	Website	13 weeks	✓	Self-administered survey at multiple time points	Positive INT had eight times higher odds of conducting thorough SSE	Participants who completed all modules were significantly more worried about recurrence[g]
Marek et al. (2018a)	The United States	RCT study	71, 41%	47[e]	NS	High risk group requiring the use of professional TBP	Smartphone application only	7 months	✓	Self-administered survey at pre- and post-intervention	Positive SSE rates increased significantly (OR = 3.39)[g]	NIL
Marek et al. (2018b)	The United States	RCT study	69, 36%	54[e]	NS	Visiting the clinic/study site and undergone TBP	Smartphone application only	6 months	✓	Self-administered survey at pre- and post-intervention	Positive SSE rates increased significantly (OR = 2.64)[g]	NIL
Murchie et al. (2015)	The United Kingdom	ASICA[c] Pilot qualitative study	20, 50%	(41–75)[e]	Yes	Treated for cutaneous melanoma[f]	Smartphone application only	6 months	✓	Self-administered survey after intervention and app data	Positive Increased proportion of participants indicating an intention to check their skin at least monthly	No change in anxiety or worry

(continued)

Table 1. (continued)

Author, year	Country	Trial and study type	Sample size (*n*), % male	Mean age or range	Includes rural or remote region	Melanoma risk/ skin cancer risk	Intervention format	Period	Includes training	Outcome measure tool/ methods	Overall SSE behavioural outcome[a]	Health outcome
Murchie et al. (2022)	The United Kingdom	ASICA[c] Pilot RCT study	240, 45%	58[e]	Yes	Treated for cutaneous melanoma[f]	Smartphone application and tablet	12 months	✓	Self-administered survey at multiple time points	Positive Significantly more regular total SSEs,[g] higher levels of self-efficacy in conducting total SSEs, and higher levels of intention to perform total SSEs	No significant difference for worry. Significantly[g] lower anxiety and depression scores and higher QoL scores at 6 months.
Ntessalen et al. (2023)	The United Kingdom	ASICA[c] Qualitative study	27, NS	(29–86)[e]	NS	Treated for cutaneous melanoma[f]	Smartphone application and tablet	12 months	✓	App data and interviews	Positive INT reported increased SSE	NIL
Reilly et al. (2021)	The United Kingdom	ASICA[c] Qualitative study	69, 45%	58[e]	Yes	Treated for cutaneous melanoma[f]	Tablet only	12 months	✓	Self-administered survey at multiple time points and interviews	Positive Most participants reported that using the digital aid increased frequency of SSE	No change or decrease in anxiety or worry
Reilly et al. (2023)	The United Kingdom	ASICA[c] Qualitative study	22, 55%	56[e]	Yes	Treated for cutaneous melanoma[f]	Smartphone application and tablet	12 months	✓	Interviews	Positive Most users reported that they regularly conducted SSE	No change or decrease in anxiety or worry
Robinson et al. (2021)	The United States	RCT study	989, 0%	(41–54)[e]	Yes	At risk for skin cancer[j] Excludes women with history of stage III/IV melanoma	Smartphone application only	3 months	X	Self-administered survey at multiple time points and medical records	Positive SSE initiation significantly increased and intervention group was two times more likely to perform SSE[g]	No significant difference in SSE anxiety between control and intervention group
Turrisi et al. (2015)	The United States	RCT study	494, 49%	(18-≥70)[e]	NS	Have previous diagnosis of stages 0–IIB melanoma	Tablet only	12 months	✓	Self-administered survey at multiple time points	Positive SSE rates were significantly higher after 4 and 12 months	NIL

Note. CG = Control Group; CSE = Clinical Skin Examination; INT = Intervention Group; NS = Not specified; OR = Odds Ratio; QoL = Quality of Life; RR = Relative Risk; SSE = Self-Skin Examination; TBP = Total Body Photography; UV = Ultraviolet.

[a]Overall SSE behaviour outcome reflects whether the study reported a positive or negative change in any of the Technology Acceptance Model domains as summarised in Table S4 in the supplementary document. [b]MELSELF: Melanoma Self-Surveillance study. [c]ASICA: Achieving Self-Directed Integrated Cancer Aftercare study. [d]Used ASICA app to deliver SSE guide. [e]Age range or mean age was not reported. [f]Previously treated for early melanoma (stages I–IIC). [g]Statistical significance at $p < .05$. [h]Statistical significance not reported. [i]ASK: Advancing Survivors Knowledge study. [j]Risk based on family and personal history of melanoma, phenotype, UV exposure by residence. [k]mySmartCheck: Adapted from mySmartSkin.

Various tools such as Melanoma Worry Scale, Hospital Anxiety and Depression Scale, Depression, Anxiety and Stress scales (DASS-21), and EQ-5D-5L were used to assess change in anxiety, depression and health-related quality of life (Ackermann et al., 2022; Horsham et al., 2020; Manne et al., 2021; Manne, Heckman, et al., 2022; Manne, Marchetti, et al., 2022; Murchie et al., 2015; Murchie et al., 2022).

Stage 1 TAM: SSE Knowledge, perceived usefulness, perceived ease of use and attitudes.

Skin Self-Examination Knowledge

Of the six studies that assessed SSE knowledge, all demonstrated some positive change in participants' understanding of how to perform an SSE and how to identify new or changing moles (Ackermann et al., 2022; Horsham et al., 2020; Manne et al., 2021; Manne, Marchetti, et al., 2022; Reilly et al., 2023; Robinson et al., 2021). Manne, Marchetti, et al. (2022) reported that participants were more likely to score higher on multiple-choice questions testing suspicious lesions after completion of a 3-month intervention using their digital visual aid when compared with usual care (Manne, Marchetti, et al., 2022).

However, participants in some studies also reported some difficulty in discerning moles of concern to monitor or take a photo of, which affected their confidence in performing SSE (Drabarek et al., 2022; Kong et al., 2020).

Perceived Usefulness

Five studies (out of 20) demonstrated digital visual aid tools to be useful by empowering users and increasing their awareness of changes on their skin (Drabarek et al., 2022; Kong et al., 2020; Manne et al., 2021; Manne, Marchetti, et al., 2022; Murchie et al., 2015). Features that allow users to securely store photos on their personal devices and track records of concerning moles were identified as contributing to the perceived usefulness of interventions (Drabarek et al., 2022; Hampton et al., 2020). Some studies reported that the involvement of HCPs, to review the photos taken by users or conduct remote assessments, were integral to the intervention's utility (Dennis et al., 2015; Drabarek et al., 2022; Horsham et al., 2020; Kong et al., 2020). Participants noted that digital monitoring led to improvements in convenience, including reductions in commuting time, waiting periods and financial costs when accessing health care services (Dennis et al., 2015; Kong et al., 2020; Reilly et al., 2023). The interventions were also perceived to enhance health system efficiency by minimising unnecessary clinician engagement (Reilly et al., 2023).

Perceived Ease of Use

Of the 16 studies that reported elements of usability, seven reported that digital aids are user-friendly and intuitive (Dennis et al., 2015; Hampton et al., 2020; Horsham et al., 2020; Manne et al., 2021; Manne, Marchetti, et al., 2022; Marek et al., 2018a; Reilly et al., 2023). Having prior experience with technology or receiving training and demonstration on how to use these digital visual aids increased perceived ease of use and supported greater adherence (Drabarek et al., 2022; Reilly et al., 2023). Ease of use was negatively affected when studies reported technical issues ($n = 5/16$; Dennis et al., 2015; Drabarek et al., 2022; Horsham et al., 2020; Murchie et al., 2015; Reilly et al., 2021), difficulty in acquiring good quality photos for assessment ($n = 7/16$; Ackermann et al., 2022; Dennis et al., 2015; Drabarek et al., 2022; Hampton et al., 2020; Horsham et al., 2020; Murchie et al., 2015; Reilly et al., 2021), or felt they required a partner to complete SSEs ($n = 8/16$; Ackermann et al., 2022; Dennis et al., 2015; Drabarek et al., 2022; Janda et al., 2020; Kong et al., 2020; Marek et al., 2018b; Ntessalen et al., 2023; Robinson et al., 2021).

Attitudes Towards SSE and Digital Visual Aids

In 13 (out of 15) studies, participants expressed positive attitudes towards performing SSE and the use of digital aids, due to increased confidence, improved perceptions of SSE and from gaining experience using digital visual aids throughout the study (Ackermann et al., 2022; Dennis et al., 2015; Drabarek et al., 2022; Hampton et al., 2020; Kong et al., 2020; Manne et al., 2021; Manne, Marchetti, et al., 2022; Marek et al., 2018b; Murchie et al., 2015, 2022; Reilly et al., 2021, 2023; Robinson et al., 2021). In one study, participants were less accepting of digitally aided SSE following the intervention, although they acknowledged its potential benefits (Horsham et al., 2020). Negative attitudes reported were mainly due to a lack of time to perform SSE and a sense of technology overload or fatigue (Dennis et al., 2015; Drabarek et al., 2022; Robinson et al., 2021).

Stage 2 TAM: Intention to perform SSE.

All seven studies that measured SSE intention reported increased intention to perform regular SSE following completion of the interventions (Ackermann et al., 2022; Dennis et al., 2015; Drabarek et al., 2022; Horsham et al., 2020; Kong et al., 2021; Murchie et al., 2015; Murchie et al., 2022). In a study by Kong et al. (2020), 27 of 28 participants revealed they intended to continue using mobile teledermoscopy to monitor their skin lesions between doctor's visits.

Stage 3 TAM: Behavioural change; frequency and thoroughness.

All 11 studies that measured SSE behavioural change reported positive impact of digital interventions on SSE behaviour when compared to usual care, as described below. There were no studies that demonstrated a negative effect of

digital aids on SSE behaviour, by either reduced thoroughness or frequency when compared to usual care.

Change in SSE Frequency

Eight studies (out of 11) showed an increase in SSE frequency, (Ackermann et al., 2022; Dennis et al., 2015; Geller et al., 2023; Marek et al., 2018b; Murchie et al., 2022; Reilly et al., 2021; Robinson et al., 2021; Turrisi et al., 2015), with two studies attributing this to the perceived usefulness, trustworthiness and ease of use of digital aids (Ackermann et al., 2022; Robinson et al., 2021).

Change in SSE Thoroughness

Six studies (out of 11) reported improved SSE thoroughness, where participants would report conducting skin checks on more or all body areas (Dennis et al., 2015; Geller et al., 2023; Manne et al., 2021; Manne, Heckman, et al., 2022; Manne, Marchetti, et al., 2022; Murchie et al., 2022). Reilly et al. (2021) also reported that the intervention supported a more systematic approach to conducting skin checks.

Health Outcomes and Health System Implications. The main health outcomes reported were anxiety and worry towards using digital interventions and skin cancer, the number of clinic appointments for skin cancer checks and the likelihood of detecting skin cancer. Out of nine studies that measured health outcomes, seven reported decreased or no change in anxiety (Ackermann et al., 2022; Horsham et al., 2020; Murchie et al., 2015, 2022; Reilly et al., 2021, 2023; Robinson et al., 2021). Only two studies (out of nine) found an increase in anxiety due to the intervention, leading to worries about disease recurrence (Drabarek et al., 2022; Manne, Marchetti, et al., 2022b).

Three studies (out of 20) showed that following the intervention, there was an increase in the number of clinic appointments for skin cancer checks (Drabarek et al., 2022; Geller et al., 2023; Manne, Marchetti, et al., 2022). Of the seven studies which tracked the outcomes of histopathology testing of excised suspicious lesions, only three demonstrated an increased likelihood of skin cancer detection as a result of using digital aids when compared with usual care (Drabarek et al., 2022; Murchie et al., 2015; Robinson et al., 2021). One study reported that the use of teledermoscopy was less sensitive at detecting clinically suspicious lesions compared with naked eye (Janda et al., 2020).

Three studies concluded that digital visual aids may reduce the number of appointments, and thus interactions with HCPs (Reilly et al., 2021, 2023; Robinson et al., 2021), while three additional studies reported an increase in interactions or desire for interactions with HCPs (Ackermann et al., 2022; Geller et al., 2023; Murchie et al., 2022). One study reported both increased and decreased interactions with HCPs depending on the number of skin lesions (Drabarek et al., 2022). Notably, six studies (out of 20) reported that participants still wanted HCP input regardless of their confidence in performing SSE, highlighting the perceived importance of clinician involvement in skin cancer monitoring (Ackermann et al., 2022; Drabarek et al., 2022; Geller et al., 2023; Kong et al., 2020; Reilly et al., 2021, 2023).

Discussion

Digital visual aids appeared to support positive SSE-related behaviour within the key TAM constructs, with limited evidence of adverse effects on anxiety or other health outcomes. In this review, positive perceptions regarding ease of use of digital visual aids were mostly attributed to highly intuitive and navigable interfaces. Digital visual aids that are perceived as easy to use may lead to better adherence to SSE routines. Adherence to self-management habits, such as SSE, and sustained use of digital visual aids, requires significant effort and commitment from individuals (Ackermann et al., 2023). Developing a digital health intervention with an intuitive user interface helps to reduce barriers surrounding digital literacy for users (Greenhalgh et al., 2017; Huang & Yang, 2020; Koh et al., 2018). Contrarily, negative perceptions, both in this review and other studies (Koh et al., 2018), were commonly stemmed from technical difficulties and challenges in viewing or capturing images of lesions on body parts that are less accessible without assistance from a partner.

Past studies reported that less than 70% of their participants engaged in regular total SSE (Aitken et al., 2004). The use of digital visual aids may support positive SSE-related activities by providing a platform to monitor compliance, send reminders if SSE is not performed, and allow automated recognition of changes in skin lesions over time (Murchie et al., 2022; Puig & Malvehy, 2013). Our study certainly supports this by revealing overall improved SSE frequency and thoroughness following use of such aids. Digital health interventions are growing in popularity, with the number of smartphone applications developed for skin cancer alone that are available in the market growing from 39 in 2014 to 66 in 2019 (Kassianos et al., 2015; Kong et al., 2021).

Some common barriers to the use of digital visual aids, identified in this review, include lack of time, technology fatigue, and lack of familiarity in the routine use of technology. Of note, most of the included studies involved older adult participants, a population which tends to have lower digital literacy but higher risk of melanoma (Cancer Council Australia, n.d.-b; Deursen, 2020; Finkelstein et al., 2023). Nevertheless, these barriers were also reported in other studies that evaluated use of technology among patients and SSE practice (Koh et al., 2018; Walter et al., 2020). Our application of the TAM to the outcomes measured suggest pathways to overcome such barriers for most people. Although melanoma is the most common cancer diagnosed in young Australians aged 15–29 years (Cancer Council Australia, n.d.-c), this priority group remains under-represented in the current evidence

base. Few studies have evaluated digital aids with younger users, who may experience different barriers, preferences or perceptions of use.

While digital aids may positively influence SSE behaviour, their uptake may not be beneficial if they lead to adverse health outcomes, such as increased anxiety or rise in false positive diagnoses. According to a 2022 report by Melanoma Institute Australia (2022), approximately 40% of patients experience anxiety attributed to a cancer diagnosis, and anxiety and depression were identified as major psychological concerns for patients at all stages of melanoma. Seven studies in this review found that the use of digital visual aids for SSE either reduced or did not change user's anxiety and worries, compared to two studies reporting increased anxiety and worry, due to heightened awareness of skin changes and possibility of finding another melanoma (Drabarek et al., 2022; Manne, Marchetti, et al., 2022). Other studies have similarly reported that digital interventions may reduce uncertainties when performing SSE and did not have negative psychological impacts on users (Habgood et al., 2021; Walter et al., 2020). Given the diverse outcomes in previous studies, further studies are needed to clarify the impact of digital visual aids on participant anxiety and depression, and other psychosocial outcomes.

Another key observation was user preference for the direct involvement of HCPs for SSE. Digital visual aids were considered most useful when HCPs reviewed the photos of lesions or offered remote consultations and assessment. Engaging HCPs is not always feasible given workforce shortages and extensive waiting time for specialist appointments (Melanoma Institute Australia, 2022). Despite this, digital visual aids for SSE could supplement in-person care and CSE. This would not only empower patients by improving general and personal health literacy, but also possibly shift the patient–provider relationship to a patient-centric and collaborative care (Jongsma et al., 2021; Nassehi et al., 2024). Nurses and dermatologists in a previous study wanted digital health interventions that were value-added to all stakeholders, easy to use, and easy to implement in their daily routine (Reinders et al., 2024).

Most included studies did not report the locality of their patients. People living in regional or rural communities are known to experience inequities in accessing health care when compared to metropolitan counterparts (Thomas et al., 2015), and there is great potential for digital health interventions to address the disproportionate burden by reaching remote and rural dwellers who have limited access to preventive services for skin cancer (Maddukuri et al., 2021; Rustad & Lio, 2021).

Most participants in the included studies were already diagnosed or treated for melanoma or were visiting a clinic for monitoring. Therefore, they may have been more informed or experienced with skin checks and more motivated to use the digital visual aids for SSE, compared to the general public (Coroiu et al., 2020; Habgood et al., 2021; Koh et al., 2018). Very few studies focused on patients with numerous moles, which could be a significant challenge and barrier to SSE (Habgood et al., 2021). Finally, the majority of studies included in this review were assessed to be "at risk" of bias, mainly due to concerns within the outcome measurement domain, such as lack of information on how the outcome was measured, or missing data (Supplemental Table S2). There is a need for more studies to apply consistent and validated outcome measures to objectively assess the clinical outcomes of using digital visual aids, such as the number of newly diagnosed skin cancers.

External to the studies included, this review was limited by the timeframe from literature search to publication, potentially missing novel studies in inclusion. In addition, the exclusion of grey literature such as conference abstracts or posters lends itself to selection bias and may have excluded unpublished findings.

Overall, it seems promising that digital visual aids would likely be accepted by patients and could reduce the burden of melanoma surveillance. However, formal longitudinal research, overcoming the technological barriers described, integrating HCPs, and measuring associated health outcomes is warranted for such applications to make a meaningful impact. If it could be demonstrated that such aids could safely facilitate SSE between clinician visits, then their dissemination would be worthwhile. The Australian Institute of Health and Welfare (2026) has launched a National Digital Health Strategy, with the aim of improving access to such aids for all population groups. It details actions including incorporating digital aids into policy and systematic education to improve both HCP and patient technological literacy (Australian Institute of Health and Welfare, 2026). With the recommended research, the dissemination of digital visual aids could be supported by this strategy.

Conclusion

Digital visual aids can improve SSE behaviours and may also have potential to reduce anxiety surrounding skin cancer preventive care, as well as to mitigate the health care disparity between rural and metro locales. Barriers such as technical literacy and poor image quality remain important to address in order for digital visual aids to reach their potential as accessible, cost-effective tools to support the prevention of skin cancer. Further standardised research with consistent outcome measures is needed to guide integration of digital visual aids into clinical practice.

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Supplemental Material

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